Closed-Form Fisher Information for Homogeneous Weibull Series Systems with Masked Failure Causes

Alexander R. Towell
Department of Mathematics and Statistics
Southern Illinois University Edwardsville
lex@metafunctor.com
ORCID: 0000-0001-6443-9897

Abstract

We derive the expected per-observation Fisher information matrix in closed form for a series system of m independent Weibull components sharing a common shape parameter. In the rate parametrization $\lambda_j = \beta_j^{-k}$, the matrix entries involve only the Euler–Mascheroni constant γ and $\pi^2/6$; no numerical integration is required. We prove that masking—observing a candidate set of possible failure causes rather than the true cause—leaves all shape-related Fisher information unchanged. The entire information loss from masking is confined to the rate block of the Fisher information matrix, where it equals the information loss derived for the exponential special case. Monte Carlo simulation with 10^6 replications confirms all analytical entries to within 0.5% relative error.

Keywords: Fisher information · Weibull distribution · series system · competing risks · masked failure cause

1 Introduction

Series systems, in which the system fails when any component fails, arise naturally in reliability engineering. When components have independent lifetimes, the system failure time is the minimum of the component lifetimes, placing this problem within the classical competing risks framework [Crowder, 2001]. A common complication in practice is *masking*: the failed component is not directly observed, and only a candidate set of possible causes is reported [Miyakawa, 1984].

Towell [2025] developed a likelihood framework for masked series systems under three regularity conditions (C1–C3) that unify right-censoring, masking, and their interaction. Towell [2026] specialized this framework to exponential components and derived the Fisher information matrix in closed form, establishing a baseline for studying information loss under masking.

The Weibull distribution is the workhorse of reliability modeling [Lawless, 2003, Meeker and Escobar, 1998]. When all components share a common shape parameter—the *homogeneous* Weibull model—the system lifetime is itself Weibull and the failed component is independent of the failure time, yielding a tractable likelihood [Pareek et al., 2009]. Existing treatments derive score equations and compute the *observed* Fisher information numerically at the maximum likelihood estimate [Pareek et al., 2009, Kundu and Gupta, 2006]. To our knowledge, the *expected* Fisher information matrix has not been obtained in closed form for this model.

This paper derives the expected per-observation Fisher information matrix for the homogeneous Weibull series system in the rate parametrization $\lambda_j = \beta_j^{-k}$. Our contributions are:

1. **Closed-form FIM** (Theorem 1): All $(m+1)^2$ entries of the Fisher information matrix reduce to elementary functions of the Euler–Mascheroni constant γ and $\pi^2/6$. No numerical integration is required.

2. **Masking invariance** (Theorem 2): Under any masking mechanism satisfying the C2 symmetry condition of Towell [2025], the shape information I_{kk} and the shape–rate cross-information I_{k,λ_j} are unchanged. Only the rate–rate block is affected.

3. **Structural decomposition**: The masked Fisher information matrix equals the unmasked matrix with its rate block replaced by the exponential masked-data Fisher information of Towell [2026], evaluated at rates $\lambda_j = \beta_j^{-k}$.

Section 2 defines the model. Section 3 derives the Fisher information matrix. Section 4 establishes masking invariance. Section 5 reports numerical verification. Section 6 concludes.

2 Model

Consider a series system of m independent components. Component j has lifetime $T_j \sim \text{Weibull}(k, \beta_j)$ with common shape $k > 0$ and component-specific scale $\beta_j > 0$. The system fails at time $T = \min(T_1, \ldots, T_m)$, and the failed component is $K = \arg\min_j T_j$.

2.1 Rate parametrization

Define the rate parameters $\lambda_j = \beta_j^{-k}$ and the system quantities

$$\lambda_{\text{sys}} = \sum_{j=1}^{m} \lambda_j, \qquad \beta_{\text{sys}} = \lambda_{\text{sys}}^{-1/k}, \qquad w_j = \frac{\lambda_j}{\lambda_{\text{sys}}}. \tag{1}$$

The system hazard function is $h(t) = k\,\lambda_{\text{sys}}\,t^{k-1}$, additive in the λ_j. The cause probabilities w_j do not depend on k, which is the key to the masking invariance of Theorem 2.

2.2 Joint density and independence

The joint density of the system failure time T and failed component K factorizes as

$$f(t, K{=}j \mid k, \boldsymbol{\lambda}) = \underbrace{k\,\lambda_{\text{sys}}\,t^{k-1}\exp\!\big(-\lambda_{\text{sys}}\,t^{k}\big)}_{f_T(t)} \cdot \underbrace{w_j}_{P(K=j)}. \tag{2}$$

Thus T and K are independent, with $T \sim \text{Weibull}(k, \beta_{\text{sys}})$ and $K \sim \text{Categorical}(w_1, \ldots, w_m)$. This factorization, a direct consequence of the common shape, is well known in the competing risks literature [Crowder, 2001, Pareek et al., 2009].

2.3 Log-likelihood

The per-observation log-likelihood for an exact, unmasked observation $(t, K{=}j)$ is

$$\ell_1(k, \boldsymbol{\lambda}) = \log k + \log \lambda_j + (k-1)\log t - \lambda_{\text{sys}}\,t^{k}. \tag{3}$$

3 Fisher information

3.1 Score equations

Differentiating (3) with respect to the parameters $\theta = (k, \lambda_1, \ldots, \lambda_m)$ gives

$$s_k = \frac{1}{k} + (1-V)\log t, \tag{4}$$

$$s_{\lambda_\ell} = \frac{\mathbf{1}(K=\ell)}{\lambda_\ell} - t^{k}, \tag{5}$$

where $V = \lambda_{\text{sys}}\,t^{k}$. Since $T \sim \text{Weibull}(k, \beta_{\text{sys}})$, the transformation $V = \lambda_{\text{sys}}\,T^{k}$ yields $V \sim \text{Exp}(1)$, and

$$\log T = \log \beta_{\text{sys}} + \frac{\log V}{k}. \tag{6}$$

3.2 Closed-form Fisher information matrix

Theorem 1. *Let $\gamma \approx 0.5772$ denote the Euler–Mascheroni constant and define the centering constant $\mu = \log \beta_{\text{sys}} + (1-\gamma)/k$. The per-observation expected Fisher information matrix for the homogeneous Weibull series system in the parametrization $\theta = (k, \lambda_1, \ldots, \lambda_m)$ is*

$$I(\theta) = \begin{pmatrix} \mu^2 + \dfrac{\pi^2}{6k^2} & \dfrac{\mu}{\lambda_{\text{sys}}}\,\mathbf{e}^{\top} \\ \dfrac{\mu}{\lambda_{\text{sys}}}\,\mathbf{e} & D \end{pmatrix}, \tag{7}$$

where $\mathbf{e} = (1, \ldots, 1)^{\top} \in \mathbb{R}^m$ and $D = \text{diag}\big((\lambda_1\lambda_{\text{sys}})^{-1}, \ldots, (\lambda_m\lambda_{\text{sys}})^{-1}\big)$.

Proof. The second derivatives of (3) are

$$\frac{\partial^2 \ell_1}{\partial k^2} = -\frac{1}{k^2} - V(\log t)^2, \tag{8}$$

$$\frac{\partial^2 \ell_1}{\partial \lambda_\ell^2} = -\frac{\mathbf{1}(K=\ell)}{\lambda_\ell^2}, \tag{9}$$

$$\frac{\partial^2 \ell_1}{\partial \lambda_\ell \, \partial \lambda_r} = 0 \quad (\ell \neq r), \tag{10}$$

$$\frac{\partial^2 \ell_1}{\partial k \, \partial \lambda_\ell} = -t^k \log t. \tag{11}$$

We compute $I(\theta) = -E[\nabla^2 \ell_1]$ using the substitution $V = \lambda_{\text{sys}} T^k \sim \text{Exp}(1)$ and the standard moments

$$E[V \log V] = 1 - \gamma, \qquad E[V(\log V)^2] = (1-\gamma)^2 + \frac{\pi^2}{6} - 1. \tag{12}$$

Rate block. Since $P(K = \ell) = w_\ell = \lambda_\ell / \lambda_{\text{sys}}$,

$$I_{\lambda_\ell, \lambda_\ell} = \frac{w_\ell}{\lambda_\ell^2} = \frac{1}{\lambda_\ell \, \lambda_{\text{sys}}},$$

and $I_{\lambda_\ell, \lambda_r} = 0$ for $\ell \neq r$.

Cross terms. Using $T^k = V/\lambda_{\text{sys}}$ and (6),

$$I_{k,\lambda_\ell} = E[T^k \log T] = \frac{1}{\lambda_{\text{sys}}} \left[\log \beta_{\text{sys}} + \frac{1-\gamma}{k} \right] = \frac{\mu}{\lambda_{\text{sys}}}.$$

Shape entry. From (8), $I_{kk} = k^{-2} + E[V(\log T)^2]$. Expanding $(\log T)^2$ via (6) and applying (12):

$$\begin{aligned} I_{kk} &= \frac{1}{k^2} + (\log \beta_{\text{sys}})^2 + \frac{2(1-\gamma)}{k} \log \beta_{\text{sys}} + \frac{(1-\gamma)^2 + \pi^2/6 - 1}{k^2} \\ &= \left(\log \beta_{\text{sys}} + \frac{1-\gamma}{k} \right)^2 + \frac{\pi^2}{6k^2} = \mu^2 + \frac{\pi^2}{6k^2}. \end{aligned}$$

□

Remark 1 (Special cases). (i) For $m = 1$, transforming to the (k, β) parametrization via $\lambda = \beta^{-k}$ recovers the classical single-Weibull Fisher information $I_{kk}^{(k,\beta)} = [(1-\gamma)^2 + \pi^2/6] \,/\, k^2$ [Lawless, 2003].

(ii) For $k = 1$ (exponential components), $\lambda_j = 1/\beta_j$ and the rate block D reduces to the unmasked exponential Fisher information of Towell [2026].

(iii) The rate block is diagonal: information about distinct component rates is orthogonal. Each rate λ_j contributes information only through the indicator $\mathbf{1}(K{=}j)$, which identifies component j as the failure cause.

(iv) The cross terms $I_{k,\lambda_\ell} = \mu/\lambda_{\text{sys}}$ are identical for all ℓ: the shape parameter couples symmetrically to all component rates through the system failure time.

4 Masking invariance

Under the C2 masking condition of Towell [2025], the observer reports a candidate set $C \subseteq \{1, \ldots, m\}$ containing the true failed component K, rather than K itself. The conditional probability $P(C \mid K{=}j)$ depends on the masking mechanism but not on the model parameters $(k, \boldsymbol{\lambda})$.

The per-observation log-likelihood for a masked observation (t, C) is

$$\ell_1^{\text{masked}} = \log k + (k-1)\log t - \lambda_{\text{sys}}\, t^k + \log\Big(\sum_{j \in C} \lambda_j\, P(C \mid K{=}j)\Big). \tag{13}$$

Theorem 2. *Under any masking mechanism satisfying C2, the shape information I_{kk} and the shape–rate cross-information I_{k,λ_ℓ} are identical to their unmasked values. The masked Fisher information matrix has the block structure*

$$I^{\text{masked}}(\theta) = \begin{pmatrix} \mu^2 + \dfrac{\pi^2}{6k^2} & \dfrac{\mu}{\lambda_{\text{sys}}}\,\mathbf{e}^\top \\ \dfrac{\mu}{\lambda_{\text{sys}}}\,\mathbf{e} & I^{\text{exp}}_{\lambda\lambda} \end{pmatrix}, \tag{14}$$

where $I^{\text{exp}}_{\lambda\lambda}$ is the rate block of the exponential masked-data Fisher information from Towell [2026], evaluated at rates $\lambda_j = \beta_j^{-k}$.

Proof. The final term in (13) depends on $\boldsymbol{\lambda}$ but not on k. Therefore

$$\frac{\partial^2 \ell_1^{\text{masked}}}{\partial k^2} = -\frac{1}{k^2} - V(\log t)^2, \qquad \frac{\partial^2 \ell_1^{\text{masked}}}{\partial k\, \partial \lambda_\ell} = -t^k \log t,$$

which are identical to the unmasked case (8)–(11). Taking expectations yields the invariance of I_{kk} and I_{k,λ_ℓ}.

For the rate block, the Hessian of $-\lambda_{\text{sys}}\, t^k$ with respect to λ_ℓ, λ_r is zero; the rate–rate information comes entirely from the $\log(\cdot)$ term in (13), which involves only the rates $\boldsymbol{\lambda}$ and the candidate set C. Since $K \perp T$ in the homogeneous model, C is independent of T, and the distribution of C depends on $\boldsymbol{\lambda}$ only through the cause probabilities $w_j = \lambda_j/\lambda_{\text{sys}}$—exactly as in the exponential case. □

Remark 2 (Information loss). The difference

$$I^{\text{complete}} - I^{\text{masked}} = \begin{pmatrix} 0 & \mathbf{0}^\top \\ \mathbf{0} & D - I^{\text{exp}}_{\lambda\lambda} \end{pmatrix}$$

is zero everywhere except in the rate block, and positive semidefinite by the data processing inequality. Masking obscures component identity but preserves the failure time, degrading information about the individual rates while leaving shape information intact.

Remark 3 (Parametrization dependence). The invariance of Theorem 2 is specific to the rate parametrization $\lambda_j = \beta_j^{-k}$. In the $(k, \beta_1, \ldots, \beta_m)$ parametrization, the cause probabilities $w_j = \beta_j^{-k} / \sum_\ell \beta_\ell^{-k}$ depend on k, so the masking term in (13) couples to k and the invariance breaks down. The rate parametrization is natural precisely because it decouples component identity from the time scale.

5 Numerical verification

We verify Theorem 1 by Monte Carlo simulation in R [R Core Team, 2025]. For each replication, we generate a system failure time T and failed component K from the joint density (2), compute the score vector (4)–(5) and Hessian (8)–(11), and accumulate two estimators: the outer product of scores and the negative Hessian.

5.1 Primary test case

Table 1 reports results for $m = 3$ components with $k = 1.5$ and $\boldsymbol{\beta} = (100, 200, 150)$, using $n = 10^6$ replications. The derived quantities are $\boldsymbol{\lambda} = (0.001, 0.0000354, 0.000544)$, $\lambda_{\text{sys}} = 0.001578$, $\beta_{\text{sys}} = 66.51$, and $\mu = 4.47$.

Table 1: Analytical FIM entries (Theorem 1) versus Monte Carlo estimates ($n = 10^6$). Relative error is reported for nonzero entries; the off-diagonal rate entries $I_{\lambda_\ell,\lambda_r}$ ($\ell \neq r$) are analytically zero and confirmed numerically.

Entry	Analytical	MC (scores)	MC (Hessian)	Max rel. error
I_{kk}	20.22	20.20	20.22	0.10%
I_{k,λ_ℓ}	2833	2832	2834	0.04%
I_{λ_1,λ_1}	634.0	633.4	633.7	0.09%
I_{λ_2,λ_2}	17 900	17 890	17 900	0.06%
I_{λ_3,λ_3}	1165	1164	1164	0.09%

All nonzero entries agree to within 0.5% relative error across both estimators.

5.2 Special cases

Single component ($m = 1$). With $k = 2$ and $\beta = 50$, the Jacobian-transformed $I_{kk}^{(k,\beta)}$ from Theorem 1 matches the standard Weibull result $[(1-\gamma)^2 + \pi^2/6]/k^2$ to machine precision.

Exponential ($k = 1$). With $\boldsymbol{\beta} = (100, 200, 150)$, the rate block of Theorem 1 matches the unmasked exponential FIM $I_{\lambda_\ell,\lambda_\ell} = 1/(\lambda_\ell\, \lambda_{\text{sys}})$ to machine precision.

6 Discussion

We derived the expected Fisher information matrix in closed form for a homogeneous Weibull series system with masked failure causes. The result (Theorem 1) involves only the Euler–Mascheroni constant and $\pi^2/6$, extending the known single-Weibull FIM to the multi-component series setting. The masking invariance (Theorem 2) shows that masking degrades only the component identification information—the rate block—while leaving shape information untouched.

Applications. The closed-form FIM enables Cramér–Rao lower bound calculations and D-optimal experimental design for Weibull series systems without numerical integration. For mixed observation types (exact, masked, right-censored), the per-observation contributions differ in form but the exact-data FIM provides an upper bound on the information per observation.

Extensions. For right-censored observations at time c, the score contributions $s_k^{\mathrm{RC}} = -\lambda_{\mathrm{sys}}\, c^k \log c$ and $s_{\lambda_\ell}^{\mathrm{RC}} = -c^k$ are available in closed form, but the expected FIM requires specification of the censoring distribution [Escobar and Meeker, 1994]. For the heterogeneous Weibull model (distinct shapes $k_1, \ldots, k_m$), the independence $K \perp T$ fails, and closed-form expressions are unlikely; numerical computation of the observed FIM remains the practical approach [Pareek et al., 2009].

Software. The analytical FIM of Theorem 1 and the Monte Carlo verification of Section 5 are implemented in R. Source code is available from the author.

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